

Bollinger Bands under Varying Market Regimes: A Comparative Study of Breakout and Mean-Reversion Strategies in BTC/USDT

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Abstract

This study investigates the effectiveness of Bollinger Bands in cryptocurrency trading by contrasting two foundational interpretations: breakout (trend-following) and mean reversion (pullback) across multiple timeframes and market regimes using BTC/USDT data from July 2017 to March 2022. High-resolution backtesting reveals that Bollinger Bands exhibit regime dependent behavior, with each strategy's performance contingent on volatility structure and directional persistence. During the Bear Phase (2018), breakout strategies outperformed by aligning with volatility expansions, while mean-reversion signals failed amid sustained declines. In the Accumulation Phase (2018–2020), mean reversion regained limited profitability in calmer markets, though breakout setups maintained superior risk-adjusted returns. The Bull Run (2020–2021) represented the most favorable environment for both frameworks, breakout captured sustained momentum, whereas mean reversion achieved exceptional gains by exploiting short-term corrections. Conversely, the Distribution Phase (2021–2022) showed the degradation of both approaches as trend conviction weakened.

This research contributes to the literature on cryptocurrency market microstructure and algorithmic trading by: (1) reframing Bollinger Band analysis through a behavioral lens contrasting two core trading philosophies; (2) systematically evaluating their efficacy across timeframes and macro-market regimes; and (3) offering evidence-based insights into the contextual suitability of volatility-based indicators for high-noise, continuously traded markets.

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1. Introduction

Technical indicators remain fundamental to traders and quantitative analysts seeking to interpret market dynamics, identify trade opportunities, and control risk across both traditional and digital asset markets. Among these, Bollinger Bands, introduced by John Bollinger in the 1980s, stand out as a volatility-adaptive tool that reflects market expansion and contraction in real time. The indicator consists of a moving average surrounded by upper and lower bands placed typically two standard deviations apart (Bollinger, 2022). By adjusting dynamically to price fluctuations, Bollinger Bands serve as a flexible framework for assessing volatility and potential price extremes.

Unlike momentum-based oscillators such as the Relative Strength Index (RSI), Bollinger Bands respond to volatility rather than trend persistence. This property makes them particularly valuable in cryptocurrency markets, which exhibit continuous trading, high volatility, and

frequent regime shifts. Bitcoin and similar assets often alternate between prolonged consolidations and sharp directional movements, providing a fertile environment for both breakout-oriented and pullback-oriented trading approaches. Prior empirical studies demonstrate the wide applicability of Bollinger Bands across asset classes—including equities (Lento, Gradojevic, & Wright, 2007), foreign exchange (Liu, Lee, & Chiang, 2012), and commodities (Liu & Zhang, 2017)—but their specific behavioral role within crypto markets remains underexplored.

Within digital assets, recent research confirms that volatility-based indicators can yield consistent signals. Hudson and Urquhart (2021) identify Bollinger Bands among the most reliable technical tools across multiple cryptocurrencies, while Sebastião and Godinho (2021) show that incorporating Bollinger Band features into machine-learning models enhances predictive power. However, despite their empirical success, there is limited clarity about how Bollinger Bands perform under contrasting trading logics; whether they function more effectively as breakout detectors capturing volatility expansions, or as pullback indicators identifying short-term reversions toward equilibrium.

This distinction is not trivial: breakout strategies assume that price escaping beyond the upper or lower band signals the onset of a new trend, while pullback strategies interpret the same movement as a temporary deviation likely to revert toward the mean. Both philosophies rely on the same indicator yet exploit opposite interpretations of market behavior. Understanding which approach better fits the structural characteristics of Bitcoin’s market, across its cyclical phases of Accumulation, Bull Run, Distribution, and Bear Market is crucial for both practitioners and researchers aiming to design effective quantitative systems.

Accordingly, this study investigates the suitability of Bollinger Bands for breakout versus pullback trading on the BTC/USDT pair. Using historical price data spanning multiple timeframes; from intraday to daily intervals, this study systematically backtests representative strategies from each family under realistic trading conditions. Each strategy employs canonical Bollinger Band parameters without hyperparameter optimization, ensuring that observed performance differences reflect behavioral characteristics rather than fine-tuning effects. By analyzing outcomes across distinct Bitcoin market phases, this study aims to reveal when and why each approach tends to prevail.

This research contributes to the growing literature on cryptocurrency market microstructure and algorithmic trading in three ways: firstly it reframes Bollinger Band analysis through a behavioral lens contrasting two core trading philosophies, secondly it evaluates these strategies across timeframes and macro-market regimes; and lastly it offers evidence-based insights into the contextual suitability of volatility-based indicators for dynamic, high-noise markets.

The remainder of this paper is organized as follows: Section 2 reviews relevant literature on Bollinger Bands and breakout versus pullback frameworks; Section 3 details the data, strategy definitions, and backtesting methodology; Section 4 presents empirical results; and Section 5 discusses implications and concludes.

2. Literature Review

Bollinger Bands, developed by John Bollinger in the 1980s, have long been regarded as one of the most versatile and empirically validated tools in technical analysis. Structurally, the indicator consists of a simple moving average surrounded by upper and

lower bands placed a fixed number of standard deviations (typically two) away from the mean (Bollinger, 2022). The primary innovation lies in its dynamic nature: as volatility expands or contracts, the bands adjust automatically, allowing traders to assess market conditions in real time. This adaptability has led to two dominant schools of interpretation in trading practice: breakout (trend-following) and pullback (mean-reverting) strategies. While both rely on the same statistical foundation, they differ fundamentally in behavioral logic. Breakout traders view price movement beyond the bands as a continuation signal, whereas pullback traders interpret it as a temporary deviation likely to revert toward the mean.

In the equity literature, these two perspectives have been explicitly contrasted. Lento, Gradojevic, and Wright (2007) were among the first to test Bollinger Band rules distinguishing between momentum-based breakouts and contrarian reversions. Their empirical results on the Dow Jones Industrial Average and NASDAQ indicated that breakout approaches generated statistically significant positive excess returns, particularly during prolonged trending periods. Conversely, contrarian strategies outperformed in range-bound markets but suffered drawdowns during directional expansions. Hsu & Kuan (2005) reinforced these findings by demonstrating that breakout style technical trading rules, conceptually related to Bollinger Band breaches, outperformed buy-and-hold benchmarks in several emerging equity markets, especially during trending conditions. Collectively, these studies established that the market's structural state; trend versus range dictates the success of the chosen Bollinger Band interpretation.

The foreign exchange literature extended this debate. Liu, Lee, and Chiang (2012) compared Bollinger Band-based rules to random walk models across major currency pairs, confirming that both breakout and pullback logics could be profitable under distinct volatility conditions. Leung, Daouk, and Chen (2000) highlighted that currencies' mean-reverting tendencies often favored contrarian or pullback variants; buying when prices breached the lower band and selling near the upper. Yet, they also noted that during macroeconomic or liquidity shocks, breakout configurations captured large directional moves missed by mean-reversion models. Their work emphasized volatility regime dependency and the importance of aligning the chosen Bollinger Band philosophy with the underlying market microstructure. Later studies proposed adaptive versions of the indicator, adjusting standard deviation multipliers to local volatility levels, but most confirmed that breakout and pullback dynamics alternated dominance over time rather than one being universally superior.

In commodity markets, both interpretations again found empirical grounding. Liu and Zhang (2017) demonstrated that Bollinger Bands successfully identified short-term mean reversions in gold, while in crude oil, band breakouts following low volatility "squeezes" often preceded sustained price trends. Their results suggested that volatility compression and subsequent expansion are key mechanisms behind breakout profitability, whereas mean reversion trades rely on liquidity driven corrections within

established ranges. Volume based confirmation filters were shown to enhance both styles by distinguishing genuine breakouts from false ones and avoiding contrarian entries during strong trend acceleration. These findings reinforced the notion that Bollinger Bands can be effective in either direction; depending on whether volatility is expanding or stabilizing.

The emergence of cryptocurrency markets has revitalized academic interest in volatility based indicators, as their structural properties high volatility, non-stationarity, and continuous trading amplify the phenomena that Bollinger Bands are designed to capture. Hudson and Urquhart (2021) analyzed a suite of technical indicators in major cryptocurrencies and found that Bollinger Band breakouts exhibited consistent predictive power across assets and timeframes, particularly during major bull markets. In contrast, mean reversion strategies using lower band entries performed more favorably in low volatility consolidation phases. Sebastião and Godinho (2021) integrated Bollinger Band derived features into machine learning models for cryptocurrency forecasting, demonstrating that the volatility envelope retained significant explanatory value even when used as an input to complex, data driven systems. These results collectively indicate that cryptocurrencies exhibit alternating market regimes in which breakout and pullback interpretations of Bollinger Bands each demonstrate contextual superiority.

Despite this growing body of literature, direct comparative analyses of breakout versus pullback Bollinger Band strategies remain limited, especially within the cryptocurrency domain. Prior studies often examined only one variant or relied on parameter optimization rather than behavioral evaluation. Yet, the question of whether Bollinger Bands function more effectively as a trend following or mean reversion tool in crypto markets remains unresolved. This research seeks to address that gap by systematically testing canonical breakout and pullback strategies under identical conditions, using standardized Bollinger Band parameters and consistent market phase segmentation. Through this comparative lens, this study aims to clarify the functional role of Bollinger Bands in volatile, regime shifting environments such as BTC/USDT, offering empirical insight into when traders should favor continuation over reversion logics.

3. Methodology

This study employs historical BTC/USDT price data obtained from the Binance exchange, covering the period from July 17, 2017, to January 1, 2022. The dataset consists of 1-minute candlestick data, providing high-resolution granularity suitable for realistic signal generation, execution timing, and evaluation of strategy performance. Using one-minute data allows the backtesting framework to capture intraday market microstructure effects and to aggregate accurately into higher-order timeframes for multi-horizon analysis. To evaluate the behavior of Bollinger Bands across varying trading horizons, the analysis is conducted across eight timeframes: 1-minute, 5-minute, 15-minute, 30-minute, 1-hour, 6-hour, 12-hour, and daily intervals. This structure facilitates a direct comparison between short-term, high-frequency trading dynamics and longer-term trend-

oriented conditions. Each timeframe is derived through resampling of the base dataset to ensure precise temporal alignment and consistent candle formation. All strategies in this study employ the standard Bollinger Band parameters as originally proposed by John Bollinger (2001); Moving average period: 20, Standard deviation multiplier: 2.

To account for structural variations in Bitcoin’s price behavior, the BTC/USDT market is segmented into four major phases: Accumulation, Bull Run, Distribution, and Bear Market. These phases correspond to the Bitcoin halving cycle and are delineated using historically verified halving dates. Accumulation: post-bear recovery and range-bound price stabilization. Bull Run: rapid price appreciation and new all-time highs. Distribution: plateau and profit-taking phase following major rallies. Bear Market: prolonged downtrend characterized by sustained price declines. This segmentation enables performance assessment of each trading approach under distinct macro-market regimes with different volatility, trend persistence, and liquidity characteristics.

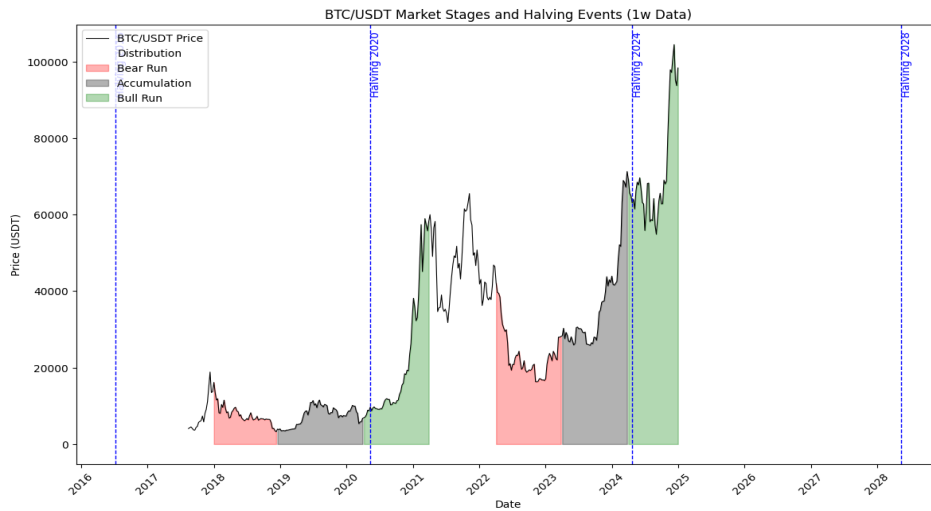


Figure 1: Bitcoin Market Cycles / Halving Dates

Two canonical Bollinger Band strategies are implemented for comparison: Breakout and Pullback.

The breakout strategy represents the trend-following interpretation of Bollinger Bands. Entry Rule: Enter a long position when the closing price exceeds the upper band, indicating a potential volatility expansion and upward breakout. Exit Rule: Close the position when the price falls below the middle band. This logic reflects the assumption that volatility expansion following band breaches signals the start of a sustained directional movement.

The pullback strategy captures the mean-reverting interpretation of Bollinger Bands. Entry Rule: Enter a long position when the closing price crosses below the lower band, implying a short-term oversold condition and the likelihood of reversion toward the mean. Exit Rule: Close the position when the price reaches or exceeds the middle band. This configuration assumes that temporary deviations from the moving average tend to self-correct within stable or range-bound markets.

Both strategies operate exclusively on the long side to ensure comparability and isolate performance differences stemming from their directional logic rather than exposure symmetry. No short-selling or leverage is employed. A comprehensive backtesting engine is developed to simulate realistic trading behavior using 1 minute timeframe detail. Each simulated trade accounts for transaction fees consistent with Binance's historical fee structures (0.075%). Trades are executed strictly on bar close to prevent look-ahead bias. For each timeframe, strategies are evaluated independently across all four market phases. To preserve analytical transparency, the study deliberately excludes stop-loss and take-profit mechanisms as well as dynamic position sizing. Each trade deploys the full notional balance, allowing for an unfiltered assessment of the intrinsic characteristics of Bollinger Band-based signal generation. This design accentuates the indicator's natural strengths and weaknesses across different volatility conditions and market regimes.

Performance is assessed using standard quantitative metrics commonly applied in algorithmic trading research: Win Rate (%): the proportion of profitable trades. Total Return: the cumulative percentage change in portfolio value across the testing period. Sharpe Ratio: average excess return per unit of total volatility. Sortino Ratio: return per unit of downside deviation, emphasizing risk-adjusted performance. Maximum Drawdown: the largest observed peak-to-trough portfolio decline. Additionally trade frequency is reported to evaluate efficiency and capital utilization and average trade duration (ATD) is presented for winning and losing trades. These metrics collectively facilitate comparison between the two strategy families under different timeframes and market regimes.

To mitigate overfitting, the analysis adopts a walk-forward validation structure, assessing strategy performance independently within each market phase. This segmentation not only reflects real-world trading cycles but also ensures that results are not dominated by a single extended trend period. Furthermore, all simulations employ identical fee and execution assumptions, guaranteeing fair comparison between the breakout and pullback frameworks.

4. Results & Analysis

During the Bear Phase of 2018 (January–December 2018), which coincided with a market-wide decline of approximately –76.50% in Bitcoin's price, both Bollinger Band strategies were subjected to extreme downward volatility and prolonged negative momentum. This period serves as a robust stress test for volatility-based systems, as persistent bearish pressure tends to invalidate mean-reversion assumptions while rewarding selective trend-following behavior.

Table 1 summarizes the performance of the Breakout and Mean Reversion strategies across multiple timeframes during this phase. Both short-term configurations (1-minute and 5-minute) suffered severe losses, reflecting the dominance of microstructure noise and transaction costs in high-frequency crypto markets. However, performance began to diverge meaningfully from the 15-minute timeframe onward. The Breakout strategy which interprets upper-band breaches as volatility expansions demonstrated growing stability at mid-range intervals. Profitability transitioned into

positive territory at the 30-minute (+5.76 %) and 1-hour (+14.89 %) levels, accompanied by markedly lower drawdowns (18.12 % and 10.18 %) and positive risk-adjusted returns (Sharpe 0.13 and 0.36). In contrast, the Mean Reversion strategy, which buys near the lower band and exits at the mid-band, remained unprofitable across all timeframes. This asymmetry underscores a structural incompatibility between dip-buying logic and sustained bearish trends. Although Mean Reversion registered a higher proportion of winning trades in certain intervals (e.g., 30 m: 106 wins vs. 78 losses), the magnitude and duration of its losing positions were substantially larger (average losing trade > 1 day 4 hours vs. winning trade $\approx$ 13 hours). The Breakout configuration exhibited the inverse pattern; fewer trades, shorter losing durations, and tighter drawdowns which is consistent with momentum alignment following volatility expansions.

In summary, during the 2018 bear cycle, Bollinger Bands proved better suited for breakout-based trend continuation strategies than for pullback entries. Mean-reversion frameworks failed to capitalize on transient oversold conditions, while breakout logic captured the limited but more durable relief rallies with superior capital preservation and risk efficiency.

Breakout Strategy Performance										Mean Reversion Strategy Performance									
Time Frame	Profit %	Number of Trades	Win	Loss	ATD Winners	ATD Losers	Max Drawdown	Sharpe Ratio	Sortino Ratio	Time Frame	Profit %	Number of Trades	Win	Loss	ATD Winners	ATD Losers	Max Drawdown	Sharpe Ratio	Sortino Ratio
1m	-99.92	3605	672	2933	43mins	20mins	99.92%	-18.81	-24.22	1m	-99.94	3365	1424	1941	31mins	1hour 3mins	99.94%	-17.80	-15.63
5m	-86.4	770	175	595	3hours 7mins	1hours 31mins	87.37%	-5.96	-10.22	5m	-85.44	1150	658	492	2 hours 29mins	5hours 42mins	87.34%	-4.39	-3.93
15m	-48.6	270	75	187	7hours 1mins	4hours 18mins	52.52%	-1.74	-3.65	15m	-53.54	387	244	143	7hours 52mins	18hours 7mins	59.66%	-1.10	-0.94
30m	5.76	122	53	51	7hours 25mins	8hours 4mins	18.12%	0.13	0.23	30m	-65.91	184	106	78	13hours 39mins	1day 14hour 48mins	71.76%	-0.95	-0.91
1h	14.89	58	31	10	7hours 46mins	14hours 48mins	10.18%	0.36	0.34	1h	-67.32	94	54	40	1day 5hours 23mins	3days 4hours 54mins	71.61%	-0.84	-0.88
6h	-9.75	8	3	2	7hours 47mins	5days 3hours	10.91%	-0.17	-0.59	6h	-67.33	16	6	10	7days 20hours	15days 3 hours 36min	67.33%	-0.40	-0.44
12h	-5.47	3	1	1	7hours 47mins	12days	6.53%	-0.09	n/a	12h	-59.08	9	4	5	8days 9 hours	33days 16hours 48min	59.08%	-0.24	-0.34
1d	-16.54	1	0	1	n/a	11days	-16.54	n/a	n/a	1d	-41.85	4	2	2	18days	65days 12hours	50.49%	-0.10	-0.15

Figure 2: Bear Market Bollinger Bands Breakout vs Mean Reversion Strategy Performance

During the Accumulation Phase (December 2018 to March 2020), during which the BTC/USDT market gained a modest 15.18 %, both strategies showed a rebound in stability and a notable reduction in drawdowns compared to the preceding Bear Phase. This period marked the transition from deep bearish sentiment to the early stages of recovery, defined by extended consolidation, lower volatility, and intermittent upward impulses.

As shown in Figure 3, both the Breakout and Mean Reversion frameworks performed poorly at ultra-short horizons (1 m and 5 m), confirming that micro level noise and fees dominate volatility based signals. Beyond the 15-minute interval, however, performance divergence became evident. The Breakout strategy began recovering at the 30-minute timeframe, where profitability turned positive (+4.87 %), improving further at 1 h (+22.10 %), 6 h (+14.16 %), and daily (+9.49 %) horizons. These intervals exhibited favorable risk metrics (Sharpe 0.07–0.36) and steadily declining drawdowns, indicating that breakout signals effectively captured early-phase volatility expansions. Balanced

win/loss distributions (e.g., 1 h: 55 wins / 32 losses) and symmetrical trade durations suggest healthy directional follow-through without excessive exposure. By contrast, the Mean Reversion strategy, which buys near the lower band and exits at the midline, remained unprofitable up to 1 h but began improving at higher timeframes. While shorter intervals still incurred losses (e.g., 30 m: -41.55 %), mid- and long-term horizons showed recovery (1 h +8.58 %, 6 h +51.87 %, Sharpe $\approx$ 0.05–0.14). This suggests that in calmer, range-bound environments, pullback entries increasingly exploited localized overextensions as volatility compressed.

Overall, the Accumulation Phase reveals a partial role reversal: breakout logic, dominant during the Bear Phase, retained superior risk-adjusted performance, while mean-reversion regained competitiveness as structural stabilization emerged. These findings confirm that Bollinger Bands' effectiveness is regime-dependent; breakouts thrive during trend formation, whereas pullbacks regain efficacy once volatility contracts and markets re-enter equilibrium.

Breakout Strategy Performance										Mean Reversion Strategy Performance									
Time Frame	Profit %	Number of Trades	Win	Loss	ATD Winners	ATD Losers	Max Drawdown	Sharpe Ratio	Sortino Ratio	Time Frame	Profit %	Number of Trades	Win	Loss	ATD Winners	ATD Losers	Max Drawdown	Sharpe Ratio	Sortino Ratio
1m	-99.94	3669	500	3169	45mins	20mins	99.95%	-23.36	-38.07	1m	-99.94	4100	1276	2824	32mins	59mins	99.94%	-24.80	-22.03
5m	-86.78	1082	248	834	3hour 25mins	1hour 31mins	89.20%	-5.13	-11.79	5m	-84.54	1467	878	589	2hour 37mins	5hour 45mins	87.91%	-4.28	-3.39
15m	-53.71	405	117	283	7hour	4hour 3mins	57.92%	-1.38	-2.51	15m	-55.29	506	327	179	7hour 37mins	17hour 17mins	68.80%	-1.14	-0.91
30m	4.87	203	88	76	7hour 27mins	7hour 1 mins	34.33%	0.07	0.09	30m	-41.55	252	163	89	14hour 50mins	1day 12hours 19mins	60.05%	-0.57	-0.49
1h	22.1	120	55	32	7hour 32mins	12hour 54mins	19.84%	0.24	0.23	1h	8.58	122	81	41	1day 9hours 7mins	2days 22hours 57mins	56.77%	0.05	0.05
6h	14.16	29	13	1	7hour 47mins	1day 18hours	6.34%	0.34	n/a	6h	51.87	25	18	7	6days 2hours 40mins	13days 12hours 51mins	29.22%	0.12	0.14
12h	5.9	14	6	0	7hour 47mins	n/a	0	0.34	n/a	12h	-31.09	9	6	3	11days	49days 16mins	40.29%	-0.09	-0.19
1d	9.49	10	4	0	7hour 47mins	n/a	n/a	0.23	n/a	1d	-20.03	5	2	3	15days	39days 16mins	40.23%	-0.04	-0.06

Figure 3: Accumulation Bollinger Bands Breakout vs Mean Reversion Strategy Performance

During the Bull Run Phase (April 1, 2020 to March 31, 2021), during which the BTC/USDT market surged by 350.46%, both strategies entered their most profitable environment, yet their behavior diverged sharply in line with their design philosophies. This period was characterized by sustained upward momentum, brief but sharp pullbacks, and heightened volatility following major macroeconomic stimulus events.

As shown in Figure 4, the Breakout strategy maintained the strongest performance at medium and higher timeframes, confirming that momentum-based Bollinger Band entries align naturally with parabolic uptrends. Profitability became positive from the 1-hour interval (+34.29 %), remaining robust through 6 h (+9.78 %), 12 h (+7.58 %), and daily (+21.17 %) horizons. Risk-adjusted returns were particularly strong between 1 h and 6 h while drawdowns declined steadily from earlier phases. These results indicate that Bollinger Band breakouts successfully captured volatility expansions aligned with directional continuation, allowing exposure to compounding trends with limited retracement risk. The Mean Reversion strategy, in contrast, achieved outstanding absolute returns at several mid-to-high timeframes, outperforming Breakout in raw profitability while maintaining controlled risk. It recorded exceptional results at 30 m (+139.04 %), 1

h (+78.41 %), 6 h (+121.03 %), and 12 h (+112.29 %), accompanied by strong Sharpe and Sortino ratios. These findings suggest that during bullish regimes, pullback-based entries near the lower band effectively exploit short term corrections within dominant upward structures. Importantly, Mean Reversion's maximum drawdowns remained low, reflecting disciplined exits near the midline that preserved profits through local consolidations.

Taken together, the Bull Run phase reveals a regime of dual viability. Breakout logic provided steady, lower-risk trend participation, whereas Mean Reversion delivered higher upside through precise re-entry on retracements. The two frameworks thus represent complementary interpretations of Bollinger Bands in strongly trending markets: Breakouts excel in continuous acceleration, while pullbacks outperform when retracement opportunities arise within a clear bullish structure.

Breakout Strategy Performance										Mean Reversion Strategy Performance									
Time Frame	Profit %	Number of Trades	Win	Loss	ATD Winners	ATD Losers	Max Drawdown	Sharpe Ratio	Sortino Ratio	Time Frame	Profit %	Number of Trades	Win	Loss	ATD Winners	ATD Losers	Max Drawdown	Sharpe Ratio	Sortino Ratio
1m	-99.94	3625	629	2996	43mins	19mins	99.95%	-34.32	-48.53	1m	-99.94	5563	2197	3366	30mins	1hour 1mins	99.94%	-36.09	-30.22
5m	-63.14	848	237	611	3hour 40mins	1hour 29mins	70.78%	-3.81	-8.64	5m	-12.78	1122	693	429	2hours 39mins	5hours 54mins	41.67%	-0.60	-0.53
15m	-23.07	330	124	201	7hour 2mins	4hour 4mins	37.45%	-0.71	-1.22	15m	77.11	380	266	114	8hours 20mins	17hours 27mins	18.77%	1.37	1.08
30m	-0.26	189	79	80	7hour 39mins	7hours 26mins	31.41%	-0.01	-0.01	30m	139.04	198	139	59	14hours 10mins	1day 10hours 40mins	14.37%	1.55	1.49
1h	34.29	124	60	23	7hour 37mins	11hour 21mins	16.20%	0.87	1.19	1h	78.41	96	69	27	1day 3hours 8mins	2days 20days 20mins	15.28%	0.93	1.07
6h	9.78	30	12	3	7hour 35mins	2days 22 hours	14.54%	0.21	0.30	6h	121.03	13	12	1	8days 11hours 30mins	11days	0.29%	0.97	n/a
12h	7.58	16	7	1	7hour 21mins	10days	9.15%	0.12	n/a	12h	112.29	9	8	1	10days 6hours	20days	5.85%	0.48	n/a
1d	21.17	13	8	0	7hour 32mins	n/a	n/a	0.58	n/a	1d	11.07	1	1	0	37days	n/a	n/a	n/a	n/a

Figure 4: Bull Run Bollinger Bands Breakout vs Mean Reversion Strategy Performance

During the Distribution Phase (April 1, 2021 to March 30, 2022), BTC/USDT declined from approximately \$59,095.81 to \$47,088.73, corresponding to a market-wide drop of around -20.3 %. This period reflected waning bullish momentum and the onset of profit-taking behavior, characterized by contracting volumes, increased intraday volatility, and extended sideways consolidation.

As shown in Figure 5, both strategies experienced performance deterioration relative to the Bull Run, but their degradation patterns differed. The Breakout strategy remained more resilient overall, maintaining profitability at the 1-hour timeframe (+13.1%), with modest drawdowns (13.27%) and slightly positive Sharpe and Sortino ratios (0.33 / 0.76). Performance at shorter intervals, however, declined sharply, indicating that the earlier momentum-driven advantage was fading as volatility lost directional structure. The consistent improvement in results toward higher timeframes suggests that breakout signals still benefitted from infrequent but decisive directional moves even in a flattening market. The Mean Reversion strategy, conversely, struggled across nearly all horizons, posting negative returns even where Breakout remained profitable. While it achieved a relatively high number of winning trades (e.g., 15 m: 250 wins / 139 losses), its cumulative PnL stayed negative (-18.49%), and drawdowns remained elevated (35-73%). Longer timeframes, such as 6 h (-23.13%) and 12 h (-

12.14%), exhibited minor stabilization but failed to recover prior gains. This pattern indicates that in markets transitioning from uptrend to equilibrium, pullback entries become increasingly unreliable as price oscillations lose depth and mean-reversion signals are prematurely triggered.

In summary, the Distribution Phase highlights a gradual erosion of both strategies' efficiency as trend conviction weakened. Breakout logic retained a marginal edge due to its ability to capture isolated directional bursts, whereas Mean Reversion underperformed under the dominance of choppy, low-amplitude movements. Together, these outcomes emphasize that Bollinger Band strategies exhibit diminishing returns in transitional, low volatility regimes; where neither breakout continuation nor dip recovery dynamics are consistently rewarded.

Breakout Strategy Performance										Mean Reversion Strategy Performance									
Time Frame	Profit %	Number of Trades	Win	Loss	ATD Winners	ATD Losers	Max Drawdown	Sharpe Ratio	Sortino Ratio	Time Frame	Profit %	Number of Trades	Win	Loss	ATD Winners	ATD Losers	Max Drawdown	Sharpe Ratio	Sortino Ratio
1m	-99.94	3201	577	2624	44mins	19mins	99.94%	-40.39	-51.21	1m	-99.94	4720	2141	2579	28mins	1hour 6mins	99.94%	-22.83	-19.68
5m	-86.24	816	209	607	3hour 17mins	1hour 26mins	86.42%	-8.05	-15.15	5m	-70.98	1156	705	451	2hours 28mins	6hours 9mins	72.59%	-3.71	-3.09
15m	-52.63	299	99	195	6hour 51mins	3hour 55mins	52.93%	-2.10	-4.46	15m	-18.49	389	250	139	8hours 2mins	18hours 51mins	35.08%	-0.43	-0.40
30m	-26.99	161	68	72	7hour 33mins	6hour 50mins	35.40%	-0.73	-1.70	30m	-40.26	192	123	69	14hours 42mins	1day 16hours 27mins	52.86%	-0.81	-0.77
1h	13.1	87	37	18	7hour 37mins	12hour 23mins	13.27%	0.33	0.76	1h	-10.72	104	66	38	1day 3hours 35mins	2days 22hours 55mins	37.20%	-0.13	-0.11
6h	-7.85	22	9	2	7hour 13mins	1day 6hours	17.50%	-0.12	-0.15	6h	-23.13	13	9	4	11days 7hours 20mins	26days 16hours 30mins	32.12%	-0.11	-0.12
12h	-8.83	8	3	2	7hour 47mins	1day 6hour	12.66%	-0.11	-0.09	12h	-12.14	8	5	3	13days 19hours 12mins	41days 16hours	23.92%	-0.05	-0.07
1d	0.6	8	2	0	7hour 47mins	-	-	0.21	-	1d	-24.77	3	1	2	15days	88days	27.84%	-0.05	-0.47

Figure 5: Distribution Bollinger Bands Breakout vs Mean Reversion Strategy Performance

5. Conclusion

This study examined the effectiveness of Bollinger Bands in BTC/USDT trading by contrasting breakout and mean-reversion interpretations across multiple timeframes and distinct market phases. The empirical results demonstrated that the indicator's performance is strongly regime-dependent, shifting according to volatility structure and directional persistence. During the Bear Phase, breakout setups outperformed by aligning with momentum-driven volatility expansions, while mean-reversion signals failed under sustained downward pressure. In the Accumulation Phase, reduced volatility allowed mean-reversion approaches to regain limited profitability, though breakout setups continued to provide steadier, lower risk returns. The Bull Run Phase marked the optimal environment for both frameworks: breakout logic captured compounding directional gains, whereas mean reversion achieved superior absolute returns by exploiting transient pullbacks. Finally, the Distribution Phase highlighted the fragility of both methods as market directionality eroded; though breakout strategies retained a marginal advantage through adaptability to sporadic volatility bursts.

This research contributes to the growing literature on cryptocurrency market microstructure and algorithmic trading in three keyways. First, it reframes Bollinger Band analysis through a behavioral lens, contrasting two foundational trading philosophies: trend-following and mean-reversion to show how identical volatility structures can elicit

opposing behavioral interpretations. Second, it evaluates these strategies systematically across timeframes and macro market regimes, revealing that profitability and stability are contingent on both the periodicity of analysis and the prevailing volatility regime. Third, it offers evidence based insights into the contextual suitability of volatility based indicators for dynamic, high noise markets such as cryptocurrencies, where structural instability and round the clock trading amplify both opportunity and risk.

Collectively, these findings underscore the importance of contextual adaptability in technical indicator design. Bollinger Bands are not inherently predictive; rather, their value emerges from how traders interpret and parameterize them relative to market conditions. Future research could extend this analysis by incorporating leverage effects, liquidity constraints, or machine learning; driven regime classifiers to dynamically toggle between breakout and mean reversion modes. Such hybrid frameworks may represent the next step toward resilient, regime aware algorithmic trading systems within the evolving landscape of digital asset markets.

6. References

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